

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2178

Stepstone 100 kV
October 31, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2178. The Interconnection Customer proposes a 178.4 MW battery energy storage facility interconnecting at the Stepstone 100 kV substation in the Indiana area. The request was received on September 09, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 700 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$486,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Concord - Segs 7 100.0 kV Ckt 1, the Segs 7 - Concord 100.0 kV Cir. 2 loads to 107.4% of its 236 MVA rating. Project DFAX: 56.3%.

The Segs 7 - Freehaven-Lakepark 100.0 kV is overloaded to 105.2% of its 266 MVA emergency rating for the loss of the Freehaven-Lakepark - Segs 7 100.0 kV #1. The project contributes 56.3% DFAX (100.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Segs 7 - Concord 100.0 kV Cir. 2	Reconductor line	\$1,042,000	26% (\$269,000)
Freehaven-Lakepark - Segs 7 100.0 kV line	Reconductor line	\$842,000	26% (\$217,000)
	Total allocated to project		\$486,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.